Definition (Semitraced symmetric stacking semicategory)
We say that a symmetric stacking semicategory $\langle \mathbf{C}, \otimes \rangle$ is *semitraced* if it is equipped with a family of *semitrace functions*

$$\mathrm{Tr}^{Z}_{X,Y} : \mathrm{Hom}_{\mathbf{C}}(X \otimes Z; Y \otimes Z) \to \mathrm{Hom}_{\mathbf{C}}(X; Y),$$

satisfying the following axioms:

1. *Compatibility with object stacking:*

   For any morphism $f : X \otimes Z \otimes U \to Y \otimes Z \otimes U$ in $\mathbf{C}$,

$$\mathrm{Tr}^{Z \otimes U}_{X,Y}(f) = \mathrm{Tr}^{Z}_{X,Y}\left(\mathrm{Tr}^{U}_{X \otimes Z, Y \otimes Z}(f)\right).$$

2. *Compatibility with morphism stacking:*

   For any morphisms $f : X \otimes Z \to Y \otimes Z$ and $g : V \to W$ in $\mathbf{C}$,

$$\mathrm{Tr}^{Z}_{V \otimes X, W \otimes Y}(g \otimes f) = g \otimes \mathrm{Tr}^{Z}_{X,Y}(f).$$

3. *Compatibility with permutations:*

   J: we need an extra axiom here I think in order to not have to define left and right traces